

# Mikhail Razakov

## Compiler / Software Protection Engineer

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Compiler/program-analysis engineer with professional LLVM/static-analysis experience and a historical Unity licensing project. Strongest inspectable signals are LLVM legality/transformations, x86-64 codegen/register allocation, and Roslyn data flow; protection experience is deliberately bounded to server-side key activation, activation limits, and Unity-client integration.

### EXPERIENCE

- 2026 — present**    **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's C#/.NET static-analysis platform.
- July — August 2026**    **MCST — Compiler Engineering Intern**
- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
  - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.
- historical project**  
• **date not claimed**    **Unity licensing prototype**
- Server-side key issuance/activation, activation-count limits, and Unity-client integration; no anti-debugging or anti-tamper claim.

### PROJECTS

#### LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv,verify, structural expectations, idempotence, and observable before/after execution comparison.

#### x86-64 Compiler Backend & Register Allocation

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

#### DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

### SKILLS

#### Compiler / LLVM

C++23, LLVM 22/IR, interprocedural analysis, legality, transformations, LLVM Verifier

#### Program Analysis

Roslyn CFG, fixed-point data flow, state propagation/joins, conservative analysis

#### Low-Level / Codegen

Rust, SysV x86-64, liveness/interference, register allocation, spills, phi lowering

#### Software Protection / Licensing

server-side activation, activation-count limits, Unity integration

### EDUCATION

HSE University — Software Engineering, 2026–2030

### RECOGNITION

#### LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

#### UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

#### MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote